

$$z = W^{[1]}x + b^{[1]}$$

$$a = \text{ReLU}(z) \longrightarrow a = g(z)$$

$$h_\theta(x) = W^{[2]}a + b^{[2]}$$

Non-linear link function

Algo 25 - TWO LAYER NETWORK PREDICT ($W^{[1]}, W^{[2]}, b^{[1]}, b^{[2]}, X$)

```
# hidden-unit = m
for i=1 to m do
    a_i ← g(w_i^{[1]} · x + b_i^{[1]}) // activation of hidden unit i
end for
return W^{[2]} · a + b^{[2]} // compute output unit
// a = [a_1 a_2 ... a_m]^T ∈ ℝ^m
```

Backpropagation → GD + chain rule

✓ optimize weights in network to minimize some objective function

<u>Linear predictor</u> $\hat{y} = W \cdot x$ <u>Objective</u> →	<u>NN non-linear predictor</u> $\hat{y} = W^{[2]} \cdot \text{ReLU}(W^{[1]} \cdot x)$
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$$\min_{W^{[1]}, W^{[2]}} \sum_n \frac{1}{2} \underbrace{\left(y_n - \sum_i W_i^{[2]} g(W_i^{[1]} \cdot x_n) \right)^2}_{e_n}$$

$L(W)$

Case 1 Gradient w.r.t $W^{[2]}$

✓ similar to linear case

$$\nabla_{W^{[2]}} = - \sum_n e_n a_n \quad \text{————— (6)}$$

Case 2 inner layer

$$L(W) = \frac{1}{2} \left(y - \sum_i w_i^{[2]} \cdot g(w_i^{[1]} \cdot x) \right)^2$$

$$\frac{\partial}{\partial w_i^{[2]}} = \frac{\partial L}{\partial g_i} \cdot \frac{\partial g_i}{\partial w_i^{[2]}} = \nabla_{w_i^{[2]}} \quad // \text{chain rule}$$

$$\frac{\partial L}{\partial g_i} = - \underbrace{\left(y - \sum_i w_i^{[2]} g(w_i^{[1]} \cdot x) \right)}_{\text{error, } e} w_i^{[2]}$$

$$= -e w_i^{[2]}$$

$$\frac{\partial g_i}{\partial w_i^{[1]}} = g'(w_i^{[1]} \cdot x) x$$

So, $\nabla_{w_i^{[1]}} = -e w_i^{[2]} \underbrace{g'(w_i^{[1]} \cdot x) x}_{z_i} \quad \text{————— (11)}$

Algo 26: TWO LAYER NETWORK TRAIN ($D, \alpha, M, \text{MaxIter}$)

```
1: W^{[1]} ← D × M matrix of small random values
2: W^{[2]} ← M-vector of small random values
3: for iter = 1 ... MaxIter do
    G ← D × M matrix of zeros
    g ← M-vector of zeros
    for all (X, y) ∈ D do
        for i = 1 to M do
            z_i ← w_i^{[1]} · X
            a_i ← tanh(z_i) // g is tanh
        end for
        ŷ ← W^{[2]} · a
        e ← y - ŷ
        g ← g - e · a // from eq. 6
        for i = 1 to M do
            G_i ← G_i - e w_i^{[2]} (1 - tanh^2(z_i)) · x // from eq. 11
        end for
    end for
    W^{[1]} ← W^{[1]} - α · G
    W^{[2]} ← W^{[2]} - α · g
20: end for
21: return W^{[1]}, W^{[2]}
```

